

# Implementation of a ZigBee Based Network for WBAN

Mahammad Firose Shaik

EIE Department

V R Siddhartha Engineering College

Vijayawada, India.

mahammadfirose.shaik2015@vit.ac.i

n

M. Monica Subashini

School of Electrical

Engineering Vellore Institute of

Technology Vellore, India.

monicasubashini.m@vit.ac.in

N Swathi

EIE Department

V R Siddhartha Engineering

College Vijayawada, India.

nadipineniswathi@gmail.com

**Abstract**— Wireless body area network (WBAN) consists of compact, low-power and tiny sensor network (SNs). They can keep patients under constant surveillance. Monitoring systems are therefore becoming increasingly important as the rapidly increasing global aging population leads to a higher for care. Monitoring the condition of patients is a significant issue in any hospital to ensure that patients are in good and stable situation. wireless technologies are used by these systems for transporting the vital signs of medical assessment. The communication of vital signs in care homes and hospitals are generally performed wirelessly. Vital signs are classified into emergency messages and information collected on a regular basis. In this ZigBee-based network, emergency messages are delivered through a relatively high baud rate and frequent messages are transmitted through a lower baud rate so that this type of network development can extend the percentage of the entire network life time and the data rate increases and the energy consumption of the network decreases.

**Keywords**—WBAN, Zigbee, Wireless network, Network lifetime, Baud rate, ECG, EEG.

## I. INTRODUCTION

Wireless Body Area Networks (WBANs) is one of the most recent approaches to healthcare identification, management and also a key stepping stone for long term intentional networks and the Internet. Health informatics in a hospital environment has become increasingly complex, with multiple parameters (ECG, EEG & temperature, etc.) to be measured and the need for a network that gathers data from many patients for analysis at a monitoring station[5]. As shown in Fig 1. At each visit, a certain health factor of each patient is used and is known as vital signs. Human vitals are including heart rate, human body temperature, oxygen saturation, blood pressure, and respiratory rate [1].

In general, temperature, heart rate, and blood pressure are the most regularly monitored wireless technology that has extremely surfaced as one of the most challenging systems in the healthcare sector. Monitoring the condition of patients is a significant issue in any hospital to ensure that the patients are in good and stable condition [2]. Our idea is to make a having to work approach that integrates sensors to measure parameters such as body temperature, heart rate, and to transfer it to a computer so that the health of the patient can be analyzed by doctors in any part of the hospital. It reduces the workload of the doctor and also order of accuracy.

Furthermore, this system utilizes the Zigbee technology, that allows all parameters to be interpreted on PC [3]. A micro-controller board is used to analyze patient input data. In addition, all conventional methods can be documented on

the computer within an interval that can be selected by the user. This paper provides healthcare providers with a focus on maximizing the performance and depth of knowledge of healthcare services by controlling costs [4].

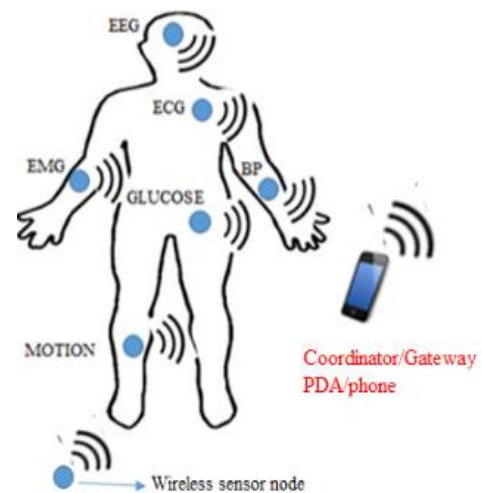


Fig 1. Basic WBAN

## II. RELATED WORK

BAN is a network that promotes all environmental factors, whether hospital, information sharing network, etc. In WBAN, patients do not need to stay a bit longer in the hospital. All essential health data will be forwarded to the physician with the support of zig-bee equipment. Routing Protocols perform an important role in the transmission of data from patient to physician [6]. Three routing protocols, that is GRP, DSDV, and AODV are used with ZigBee technology. In [7], Author provided an overview of WBAN, difficulties and implementations with open research problems that provide the required information for future research topics.

This paper presents a low energy consumption ZigBee device for ZigBee-based wireless monitoring of patients. In the case of a movable sensor network, the latest scheme considers the nearest data sink as the location in WBAN. The delay of the data path query and the controlled messages number can thus be decreased concurrently [8].

Even though numerous ZigBee devices are near the area, communication is accepted for only one ZigBee device. So, the access switch should be enforced since there can be plenty of ZigBee strategies in the vicinity. [9] proposed a mechanism for group ID to enforce network access. Physiological data shouldn't be vanished, but

wireless communication, like ZigBee, may end up losing data.

### III. ZIGBEE SYSTEM

In many cases, it is in need of continuous and long-term tracking of the vital parameters of the patient. Even so, patients are often set to release too early due to the use of a hospital bed for yet another patient on the waiting list who requires to be hospitalized immediately or because of the patient's financial inability. Our work's main aim is sketching an automatic system for monitoring patients' vital parameters, such as heartbeat and temperature, cost-effectively. We have the work divided into two parts, the transmitter and the receiver.

The Zigbee device helps with the wireless communication and the microcontroller present in the ARDUINO [2] board system signals and needs to send them for wireless communication. There are two sensors which are used to measure vital parameters. They're the heartbeat sensor and the temperature sensor.

#### A. ZIGBEE TRANSMITTER

Define abbreviations and acronyms the first time they are used. The output of the temperature sensor and the heartbeat sensor is given to the Arduino board microcontroller. The Arduino board is connected to the ZigBee transmission line as in Fig 2 for the wireless communication of data.

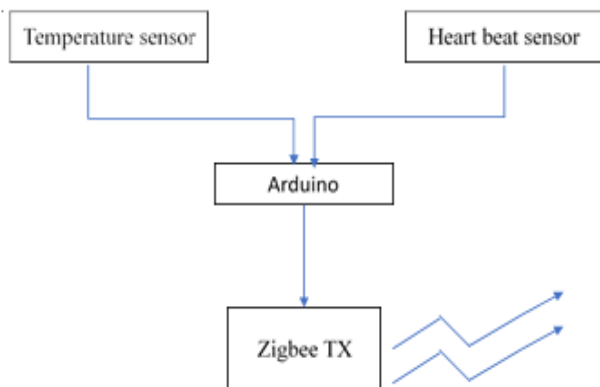


Fig 2. Zigbee transmitter

#### B. RECEIVER

The data is received from the TX antenna by the RX antenna. This signal is given as input to the ATMEGA328 microcontroller[2] of the Arduino board, which then processes it and transforms it to digital values. These extracted features are shown on the LCD screen. As shown in Fig 3.

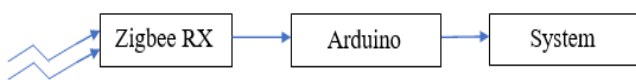


Fig 3. Zigbee transmitter

### C. ZIGBEE DEVICES

ZigBee, a low distinct wireless communication innovation which established fastly in recent years. The Zigbee Alliance and the IEEE802.15.4 team are set up together with a number of companies, stressing low costs and low energy consumption. An explosive growth of ZigBee leads for the acceptance of features that aren't limited in general personal devices, but also introduced in industry and daily life. Fig 4 is a low transmission rate Zigbee (250 Kbps), short distances (usually 50m-100m) and higher efficiency is easily placed, equipped and hold up a huge set of network nodes and several networking protocols. This also works fast, safe and reliable with low costs.



Fig 4. Zigbee Board

The amount of transmission of ZigBee is relatively lower than that utilized for effortless wireless control, such as usage for domestic industries and medicines, poison gas (carbon monoxide) identification, medical sensors and patient emergency alarms. This promotes treatment processes through transferring signal processing at any time compared to conventional wired equipment. The placement component of wireless networks can continuously monitor movement patterns and body temperature or pulse simultaneously.

Zigbee wireless communication configuration is often chosen for healthcare processes that support frequent measurement results and message transmission of data that require low energy consumption. In addition, given the simplicity of operation and acceptance of users, ZigBee, being easy laid at low cost and supports different networks. One of the node among all other can be chosen as the signal communication channel to make continuous functioning of the power system, an important security for the senior, Whenever a node is failed.

Table 1 shows that Bluetooth, while comparable in features to ZigBee, doesn't really offer a range of configurations, and its backup current is almost 70 significantly greater than ZigBee. Of the three wireless networks to be compared, ZigBee is really the only one offering mesh topology.

Table 1 Bluetooth, Wi-Fi and Zigbee characteristics

| Wireless Parameter  | Wireless Standard Devices       |              |               |
|---------------------|---------------------------------|--------------|---------------|
|                     | ZigBee                          | Wi-Fi        | Bluetooth     |
| Frequency band      | 2.4 GHz                         | 2.4 GHz      | 2.4GHz        |
| Physical/MAC layers | IEEE 802.15.4                   | IEEE 802.11b | IEEE 802.15.1 |
| Range               | Indoors: up to 30 m<br>Outdoors | 75m to 90m   | 9m            |
| Raw data rate       | 250kbps                         | 11mbps       | 1mbps         |

ZigBee, the only one offering mesh topology, is the three wireless networks in comparison. In addition, the ZigBee Destination Node can be in sleep state and still be linked to its network. ZigBee is regarded a more advanced network especially in comparison to either Bluetooth or Wi-Fi.

#### IV. BAUDRATE OF ZIGBEE DEVICES FOR WBAN

The baud rate is the rate at which the data is communicated to the communication channel. A number related to the transmission speed of data in the system. The response reflects the number of electrical fluctuations per second taking place during the transmission of data. The greater the baud rate, the additional bits per second are transported. In the synchronous serial context, the term "9600baud" implies that the serial port is capable of transmitting a maximum of 9600 operations per second. Table 2 is the BD setting Baud rate, with conventional baud rates characterized as codes 0 to 7.

Table 2 Bluetooth, Wi-Fi and Zigbee characteristics

| Bud rates defined as the codes | Baud rate                 |
|--------------------------------|---------------------------|
| 0x0000                         | 1200 bps                  |
| 0x0001                         | 2400 bps                  |
| 0x0002                         | 4800 bps                  |
| 0x0003                         | 9600 bps – device Default |
| 0x0004                         | 19200 bps                 |
| 0x0005                         | 38400 bps                 |
| 0x0006                         | 57600 bps                 |
| 0x0007                         | 115200 bps                |

The cable length will need to be reduced for wired connections at baud rates above 76,800. The higher the baud rate, the further sensitive the cord is to the reliability of installation, such as how much wire is unwrapped around each device. A number linked to the speed of transfer of data in the database. The response reflects the number of electrical oscillations per second occurring during the data transmission. The complex the baud rate, more and more bits to be transferred per sec.

Here are the USB Breakout boards to host the Tarang modules. Future based On digital technology International Ltd are used for these formation. IC FT232 is designed to supply the USB to UART features. These USB Breakouts don't need an exterior power supply (USB powered) or a handy one. One among the Tarang modules positioned in the USB Arduino microcontroller, linked with the USB port of computer the other to same port as shown in Fig 5.

Now open the software of the TMFT (Tarang Multi-Function Tool). This will repeatedly adjust the ports where devices are paired as shown in Fig 6.

To confirm the presence of the modem, select the first modem port and click the "Request modem" button. For opening the Terminal window, click CTRL+T. The command mode can be entered by using 3 continuous +++ symbols, which is confirmed with OK by modem Unlike the first XBEEs, Taranga will stay in the same command mode until the ATGEX command is received.

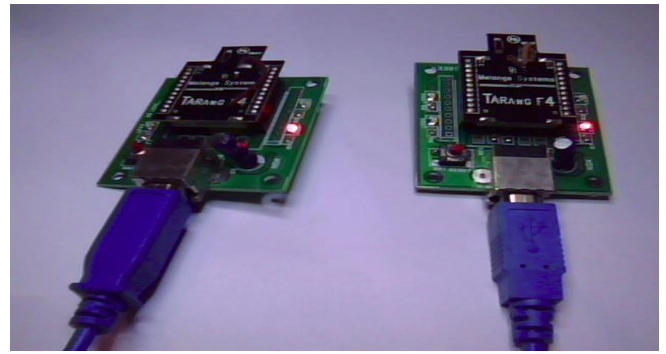


Fig 5. Zigbee transmitter

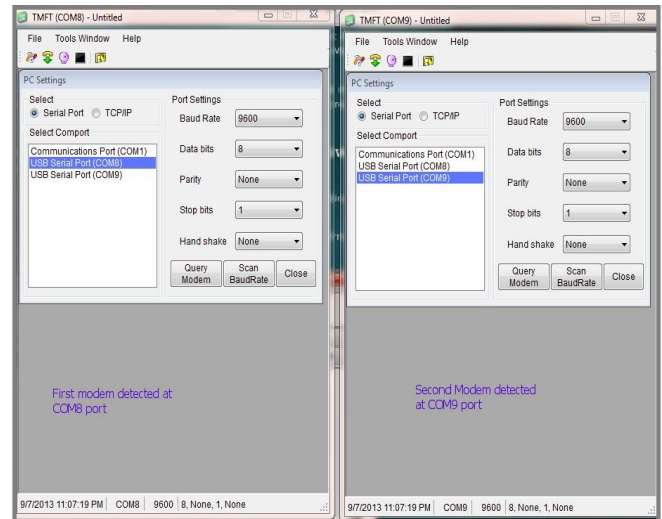


Fig 6. Zigbee transmitter

Type the defaults in the ATGRD command to Reset to factory. Don't ignore to release an ATGWR instruction to write adjustments to the modem storage. ATSD, an instruction to study the router baud rate. Getting a response from Fig 7 suggesting that the baud rate is 9600. In ATBD4 it can be changed as 9200 baud rate. The Baud modem rate is changed now.

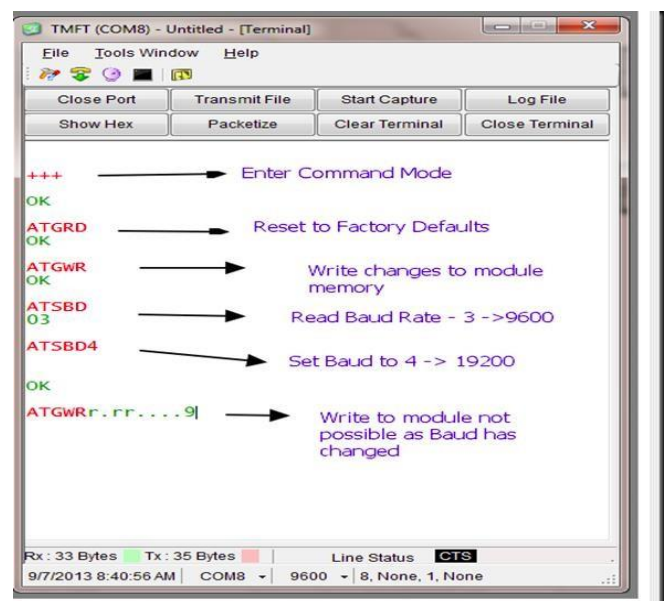


Fig 7. Zigbee transmitter

For accepting a new command, the Bluetooth terminal module must pair with that the modem. Select the below arrow at the end of the terminal close to 9600 Fig 8. Select a new Baud rate 19200 from the pop-up list.

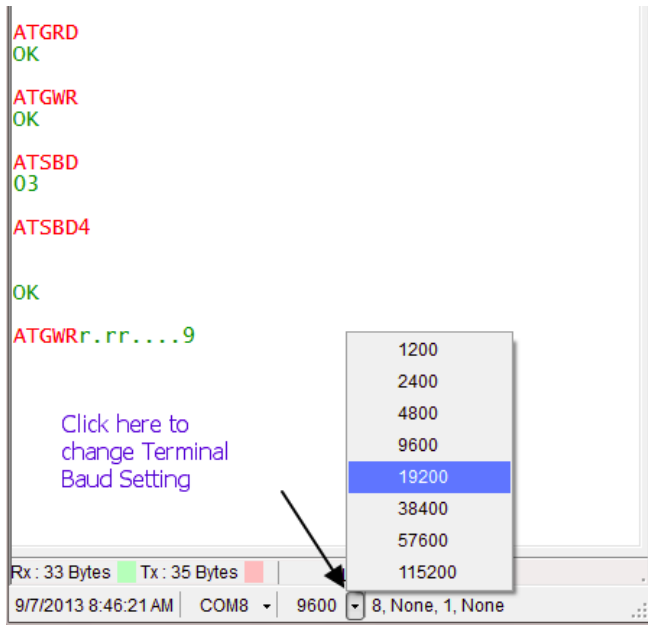


Fig 8. Changing of the Baud rates

Later, if typed in the ATGWR command, the improvements is documented in the modem. For checking the type in ATSD to see 04, that would have been the latest baud rate of 19200 in Fig 9 To exit command mode, enter the ATGEX command.

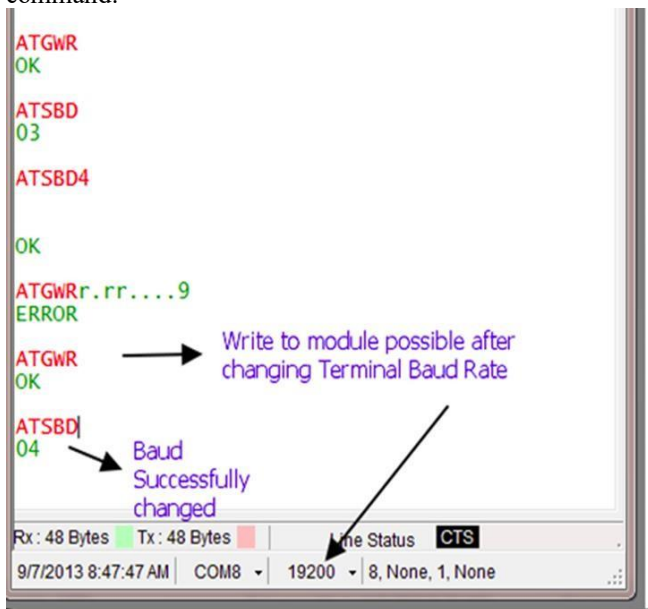


Fig 9. Changing of the Baud rates

Similarly, the baud rate is chosen for all other modem as 19200. The conversation between multiple tarang modules is now restored at a new baud rate. Use the quickest bit-rate you can afford, so that the transceiver does not have to be switched on longer than is necessary. Using lower bit rates does not make the slightest difference in power, but in energy, as it takes longer to transfer.

It also helps to use a single transmission to send 100 kB, for example, than to switch the device 10 times on and off to send a 10 kB block each time. That's because the transmitter has a non-negligible set time, and if you convey it in 10 blocks it would disperse power in 10 times that proposed platform, otherwise only once.

## V. RESULTS AND DISCUSSION

A low-cost personal healthcare monitoring implementation using a wireless sensor network to aid personal and patient-centered health care management is studied. Complex physiological data from the sensors such as the heart rate sensor Fig 10. As well as the temperature sensor are monitored and evaluated in the system and the results are displayed on the LCD and the monitor.

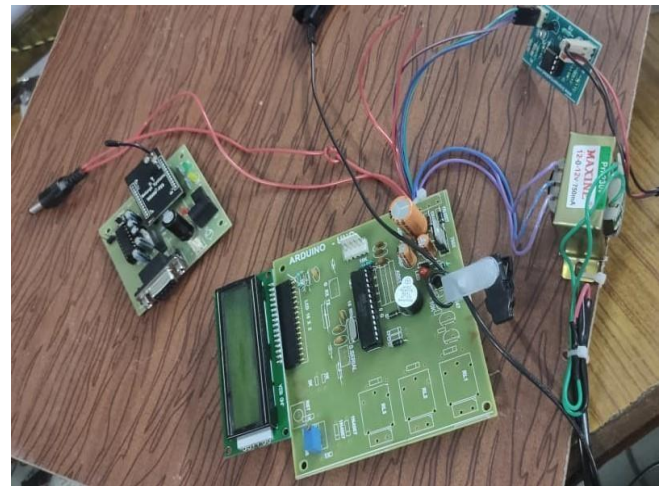


Fig 10. Changing of the Baud rates

The sensors used are rechargeable batteries and use wireless communication, this system can benefit from intelligent sensor management as well as provide a means to trade between data quality, power efficiency, and latency. The information is moved to Zigbee Devices. The information is transferred to Zigbee Devices. Both abnormal and normal data would be sent to the monitor and to the LCD Fig 11. The receiver segment is placed at the level of the staff so that they can check the situation of the person and take appropriate action.



Fig 11. Changing of the Baud rates

By using different baud rates, the transceiver would not have to be switched on longer than is necessary. Using lower bit rates can create a minimal difference in power Fig 12.

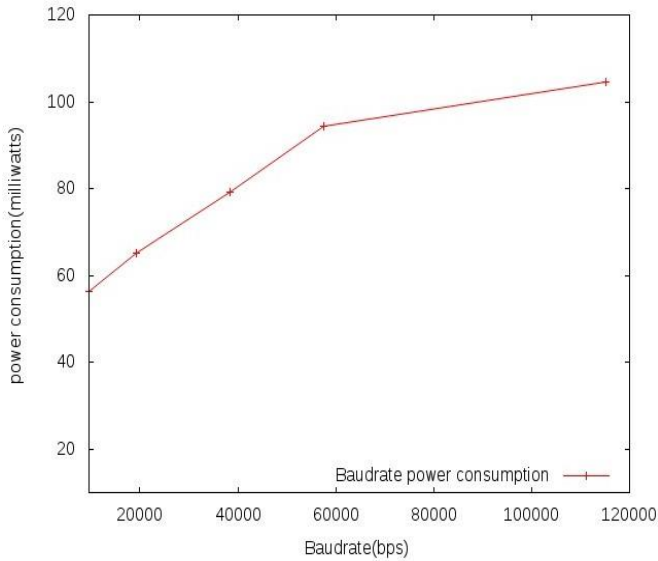


Fig 12. Changing of the Baud rates

Changing baud rate improving in power consumption. For WBAN needs low bit transfer rate to high bit transfer rate.

#### CONCLUSION

With the rapid development process of urbanization and industrialization in the world, and therefore with the increase in people's hectic schedule, it's become difficult for constant monitoring of the patient's condition. In order to avoid unexpected, hold a constant eye on the patient's condition. In order to avoid unexpected health problems and to obtain greater accuracy for diagnosing the patient's health conditions, integrated and effective data collection, control and monitoring makes a pivotal role in improving the health care system and making it more reliable and effective.

#### REFERENCES

- [1] Yuce M.R.: Implementation of wireless body area networks for healthcare systems. *Sensors and Actuators A: Physical*. 162(1):116-29 (2010).
- [2] M. M. Alam, E. Ben Hamida.: Surveying wearable human assistive technology for life and safety-critical applications: Standards, challenges and opportunities, *Sensors*, 14(5), 9153–9209 (2014).
- [3] Cavallari, R., Martelli, F., Rosini, R., C. Buratti., & R. Verdone: A Survey on Wireless Body Area Networks: Technologies and Design Challenges. In *IEEE Communications Survey & Tutorials*, 16(3), 1635-1657.
- [4] Darwish A, Hassanien AE. Correction: Darwish, A., and Hassanien.: AE Wearable and Implantable Wireless Sensor Network Solutions for Healthcare Monitoring. *Sensors* 2011, 11, 5561-5595. *Sensors*. 12, 12(9):12375-6 (2012).
- [5] H. Anandakumar and K. Umamaheswari, "A bio-inspired swarm intelligence technique for social aware cognitive radio handovers," *Computers & Electrical Engineering*, vol. 71, pp. 925–937, Oct. 2018. doi:10.1016/j.compeleceng.2017.09.016
- [6] Anupam Jain & Minakshi Halder: An Overview of Wireless Body Area Network (WBAN) using Zigbee Technology *International Journal of Scientific Development and Research* (2016), 1(5), 888-895.
- [7] Shashwat Pathaka, Mayur Kumara, Amrita Mohanb, and Basant Kumara: Energy Optimization of ZigBee based WBAN for Patient

Monitoring. In 4<sup>th</sup> International Conference on Eco-friendly Computing and Communication Systems

- [8] S. Chen et al: A Reliable Transmission Protocol for Zigbee-Based Wireless Patient Monitoring, in *IEEE Transactions on information Technology in Biomedicine*, 16(1), 6-16.
- [9] J. Jung and J. Lee, "ZigBee Device Access Control and Reliable Data Transmission in ZigBee Based Health Monitoring System," *2008 10th International Conference on Advanced Communication Technology*, Gangwon-Do, 2008, pp. 795-797, doi: 10.1109/ICACT.2008. 4493.